ESAT Mathematics 1 Formula + Fact Sheet

Compact but comprehensive recall sheet for the Mathematics 1 module. Built for no-calculator, multiple-choice ESAT practice.

Exam facts: 27 multiple-choice questions, 40 minutes, separately timed, no calculator, 1 mark per correct answer and no negative marking.

SI PREFIXES

Prefix	Power	Prefix	Power
nano	10^{-9}	deci	10^{-1}
micro	10^{-6}	kilo	10^{3}
milli	10^{-3}	mega	10^{6}
centi	10^{-2}	giga	10^{9}

Negative indices in units are expected, e.g. $\mathrm{m\,s^{-1}}$

COMPOUND UNITS

$$\text{speed} = \frac{\text{distance}}{\text{time}}$$
$$\text{density} = \frac{\text{mass}}{\text{volume}}$$
$$\text{pressure} = \frac{\text{force}}{\text{area}}$$
$$\text{unit price} = \frac{\text{cost}}{\text{quantity}}$$

Convert units before substituting.

OPERATIONS AND PRIORITY

Use brackets, powers, roots and reciprocals in the correct order.

Cancel common factors, not common terms.

$\frac{a+b}{a} \neq b$, but $\frac{a(1+b)}{a} = 1+b$, if $a \neq 0$.

FACTORS, MULTIPLES AND PRIMES

HCF: common prime factors with smaller powers.

LCM: all prime factors with larger powers.

Every integer greater than 1 has a unique prime factorisation.

INDEX LAWS

$$a^m a^n = a^{m+n}, \quad \frac{a^m}{a^n} = a^{m-n}$$
$$(a^m)^n = a^{mn}, \quad (ab)^n = a^n b^n$$
$$a^{-n} = \frac{1}{a^n}, \quad a^{1/n} = \sqrt[n]{a}, \quad a^{m/n} = \sqrt[n]{a^m}$$

STANDARD FORM

$a \times 10^n$, where $1 \leq a < 10$ and $n \in \mathbb{Z}$.

Multiply: multiply front numbers and add powers.

Divide: divide front numbers and subtract powers.

SURDS AND EXACT FORM

$$\sqrt{ab} = \sqrt{a}\sqrt{b}, \quad \sqrt{a^2 b} = |a|\sqrt{b}$$

For $a > 0$, $\sqrt{a^2 b} = a\sqrt{b}$.

$$(a+\sqrt{b})(a-\sqrt{b}) = a^2 - b$$

Rationalise using the conjugate.

RECURRING DECIMALS + ESTIMATES

Let x be the recurring decimal; multiply by $10, 100, 1000, \ldots$, align, then subtract.

Estimate first; use powers of 10, simple fractions and $\pi \approx 3.14$ or 3.

Check exact form, decimals and significant figures.

Ratio, proportion, percentages and bounds

High-frequency no-calculator tools: ratio sharing, percentage change, proportion, scale factors and error intervals.

RATIO NOTATION AND SHARING

For ratio $a : b$, total parts $= a + b$.

First share $= \dfrac{a}{a+b} \times \text{total}$.

Second share $= \dfrac{b}{a+b} \times \text{total}$.

A ratio can be a multiplicative comparison or a fraction of a whole.

SCALE FACTORS AND SIMILARITY

Length scale factor: k.

Area scale factor: k^2.

Volume scale factor: k^3.

Map scale: convert units before comparing.

DIRECT AND INVERSE PROPORTION

$y \propto x^n \Rightarrow y = kx^n$.

$y \propto \dfrac{1}{x^n} \Rightarrow y = \dfrac{k}{x^n}$.

Find k from one known pair. Direct graphs pass through origin; inverse graphs are reciprocal-shaped.

PERCENTAGES

$$\% \text{ change} = \frac{\text{new} - \text{old}}{\text{old}} \times 100.$$

Increase by $r\%$: multiply by $1 + \dfrac{r}{100}$.

Decrease by $r\%$: multiply by $1 - \dfrac{r}{100}$.

Original value: divide by the multiplier.

GROWTH, DECAY AND INTEREST

Compound growth: $A = P(1+r)^n$.

Compound decay: $A = P(1-r)^n$.

Simple interest: $I = Prt$, with r as a decimal.

Iterative process: apply the same update repeatedly.

BOUNDS AND ERROR INTERVALS

Nearest unit: $x - 0.5 \le T < x + 0.5$.

Nearest tenth: $x - 0.05 \le T < x + 0.05$.

Nearest hundredth: $x - 0.005 \le T < x + 0.005$.

For truncation, the interval is one-sided.

SYSTEMATIC COUNTING

If task 1 has m choices and task 2 has n choices, ordered choices $= mn$.

Use lists, grids, tables and tree diagrams to avoid double counting.

Check whether repetition is allowed and whether order matters.

FRACTIONS AND PERCENTAGES

$\dfrac{a}{b}$ of a quantity means multiply by $\dfrac{a}{b}$.

$p\% = \dfrac{p}{100}$.

Percentages greater than 100% are allowed.

Convert freely between fractions, decimals and percentages.

Algebra, quadratics, inequalities and sequences

Core algebra for Mathematics 1: expressions, identities, quadratics, simultaneous equations, inequalities and sequences.

ALGEBRA VOCABULARY

Expression: no equals sign. Equation: solve for values.

Identity: true for all allowed values.

Formula: connects quantities; rearrange using inverse operations.

Factor before cancelling; cancelled denominators still cannot be zero.

EXPANSION AND FACTORISATION

Collect like terms carefully.

$a^2 - b^2 = (a-b)(a+b)$.

$x^2 + bx + c = (x+p)(x+q)$ if $p+q=b$, $pq=c$.

For $ax^2 + bx + c$, look for factor pairs after expansion check.

QUADRATIC FORMULA

For $ax^2 + bx + c = 0$:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Discriminant: $\Delta = b^2 - 4ac$.

$\Delta > 0$: two real; $\Delta = 0$: repeated; $\Delta < 0$: none.

COMPLETING THE SQUARE

$$x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 + c - \frac{b^2}{4}.$$

Vertex of $y = ax^2 + bx + c$: $x = -\dfrac{b}{2a}$.

Use for turning points and minimum/maximum values.

FORMULA REARRANGEMENT

Do inverse operations in reverse order.

Clear fractions early if helpful.

Keep brackets around negative expressions.

Check by substituting a simple value into both forms.

LINEAR INEQUALITIES

If multiplying/dividing by a negative, reverse the sign.

Represent solutions on a number line, graph, or in words.

For two variables, shade the correct side of the boundary line.

SIMULTANEOUS EQUATIONS

Linear/linear: eliminate or substitute.

Linear/quadratic: substitute the linear expression into the quadratic.

Solutions are intersection points of graphs.

Always back-substitute to find both coordinates.

SEQUENCES

Linear sequence: constant first difference.

Arithmetic-type nth term: $u_n = a + (n-1)d$.

Quadratic sequence: constant second difference.

If $u_n = an^2 + bn + c$, second difference $= 2a$.

Graphs, gradients and coordinate geometry

Mathematics 1 includes graph interpretation, approximate solutions, gradients and areas under graphs, but not calculus.

STRAIGHT LINES

Gradient: $m = \dfrac{y_2 - y_1}{x_2 - x_1}$.

Line: $y = mx + c$.

Point-gradient: $y - y_1 = m(x - x_1)$.

Parallel: same gradient. Perpendicular non-vertical lines: $m_1 m_2 = -1$.

COORDINATES

$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.

$M = \left(\dfrac{x_1 + x_2}{2}, \dfrac{y_1 + y_2}{2}\right)$.

Use all four quadrants carefully with signs.

COMMON GRAPHS

- Line: $y = mx + c$.
- Quadratic: roots, intercepts, turning point.
- Simple cubic: possible 1, 2 or 3 real roots.
- Reciprocal: $y = \dfrac{1}{x},\ x \neq 0$.
- Exponential: $y = k^x,\ k > 0$.
- Trig: $\sin x, \cos x, \tan x$ in degrees.

GRAPH INTERPRETATION

Distance-time gradient = speed.

Speed-time area = distance.

Speed-time gradient = acceleration.

Use graphs for approximate solutions and context interpretation.

AREAS UNDER GRAPHS

Use geometric areas or trapezia from the graph.

Area under a speed-time graph gives distance travelled.

Area can be estimated even for non-linear graphs.

This is not calculus in Mathematics 1.

QUADRATIC GRAPH FACTS

Roots are x-intercepts.

c is the y-intercept in $ax^2 + bx + c$.

Turning point from completing the square.

$a > 0$: opens upward; $a < 0$: opens downward.

APPROXIMATE SOLUTIONS

Intersections of graphs correspond to solutions of simultaneous equations.

Approximate roots are where the graph crosses the x-axis.

Do not over-read exact values from a sketch.

DIRECT/INVERSE PROPORTION GRAPHS

Direct proportion: straight line through the origin.

Inverse proportion: reciprocal-type curve.

In contexts, check axes and units before interpreting gradients or areas.

Geometry, circles, transformations and measures

Core geometry facts, shape formulae, circle theorems, transformations, bearings and scale drawings.

ANGLE FACTS

Straight line: 180°. Around a point: 360°.

Triangle sum: 180°. Exterior angle = sum of opposite interior angles.

Polygon interior sum: $(n-2)180^\circ$. Exterior angles sum to 360°.

CONGRUENCE AND QUADRILATERALS

Triangle congruence: SSS, SAS, ASA, RHS.

Parallelogram: opposite sides parallel/equal.

Rectangle: parallelogram with right angles. Square: rectangle + rhombus.

Rhombus: all sides equal. Kite: adjacent equal pairs. Trapezium: one pair of parallel sides.

CIRCLE THEOREMS

- Centre angle is twice circumference angle.
- Angle in a semicircle is 90°.
- Angles in the same segment are equal.
- Opposite angles in a cyclic quadrilateral sum to 180°.
- Tangent is perpendicular to radius.
- Alternate segment theorem.

PYTHAGORAS AND 3D

$a^2+b^2=c^2$.

3D cuboid diagonal: $d=\sqrt{l^2+w^2+h^2}$.

For 3D, first find a face diagonal, then use Pythagoras again.

AREAS AND VOLUMES

Triangle: $\frac{1}{2}bh$. Parallelogram: bh.

Trapezium: $\frac{1}{2}(a+b)h$.

Cuboid: lwh. Right prism: cross-section area $\times$ length.

Cylinder volume: $\pi r^2 h$.

CIRCLES AND SECTORS

Circle: $A=\pi r^2$, $C=2\pi r=\pi d$.

Degrees: arc $=\dfrac{\theta}{360}\times 2\pi r$.

Degrees: sector $=\dfrac{\theta}{360}\times \pi r^2$.

Segment area = sector area - triangle area.

TRANSFORMATIONS

Translation: same vector for every point.

Reflection: mirror line is perpendicular bisector of point-image segment.

Rotation: centre, angle, direction.

Enlargement: centre and scale factor; negative scale reverses through centre.

BEARINGS, PLANS AND ELEVATIONS

Bearings are measured clockwise from North.

Use three figures: 047°, 230°.

Plans/elevations are 2D views of a 3D object.

Scale drawings: convert units before using scale.

Right-triangle trigonometry and vectors

Mathematics 1 expects right-triangle trigonometry, exact values and vectors. Sine and cosine rules are not required for Mathematics 1.

Specification caution: Mathematics 1 candidates are not expected to recall or use the sine rule or cosine rule.

RIGHT-TRIANGLE TRIGONOMETRY

$$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}}.$$

$$\cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}}.$$

$$\tan\theta = \frac{\text{opposite}}{\text{adjacent}}.$$

Use only in right-angled triangles unless reducing a general triangle to right triangles.

EXACT TRIG VALUES

θ	0°	30°	45°	60°	90°
$\sin\theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\tan\theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	undefined

TRIG GRAPHS IN DEGREES

$\sin x, \cos x$: period 360°.

$\tan x$: period 180°.

Use symmetry to find values for angles of any size.

Remember $\tan x$ is undefined where $\cos x = 0$.

VECTOR OPERATIONS

$(a, b) + (c, d) = (a + c, b + d)$.

$k(a, b) = (ka, kb)$.

Column vectors use the same component rules.

VECTOR BETWEEN POINTS

From $A(x_A, y_A)$ to $B(x_B, y_B)$:

$\overrightarrow{AB} = (x_B - x_A,\ y_B - y_A)$.

Magnitude: $|(a, b)| = \sqrt{a^2 + b^2}$.

VECTOR GEOMETRY ARGUMENTS

Parallel vectors are scalar multiples: $\mathbf{a} = k\mathbf{b}$.

Midpoint vector: $\mathbf{m} = \dfrac{\mathbf{a} + \mathbf{b}}{2}$.

Use equal/parallel side vectors to prove shape properties.

SIMILARITY LINKS

Trigonometric ratios are unchanged in similar right triangles.

For enlarged shapes, angles are preserved.

Use scale factors before applying trig if a diagram is scaled.

3D TRIG STRATEGY

Draw the relevant right triangle inside the solid.

Use Pythagoras to make a hidden side if needed.

Then apply SOH-CAH-TOA.

Statistics and probability

Probability is usually represented with tables, frequency trees, Venn diagrams and tree diagrams rather than formal set notation.

DATA DISPLAYS

Categorical data: tables, bar charts, pie charts, pictograms.

Discrete numerical data: vertical line charts.

Time series: tables and line graphs.

Choose the display that matches the data type.

AVERAGES AND SPREAD

Mean: $\bar{x} = \frac{\sum x}{n}$.

Median: middle value after ordering. Mode: most frequent.

Range $= \max - \min$.

$IQR = Q_3 - Q_1$.

GROUPED DATA

Estimated mean: $\bar{x} \approx \frac{\sum fx}{\sum f}$.

Use class midpoints for x.

Modal class = class with highest frequency.

Grouped statistics are estimates.

HISTOGRAMS AND CUMULATIVE FREQUENCY

$$\text{frequency density} = \frac{\text{frequency}}{\text{class width}}.$$

Area of a histogram bar represents frequency.

Cumulative frequency graphs estimate medians and quartiles.

SCATTER GRAPHS

Positive correlation: both tend to increase together.

Negative correlation: one tends to decrease as the other increases.

Correlation does not imply causation.

Interpolation is safer than extrapolation.

PROBABILITY SCALE

$0 \le P(A) \le 1$.

Exhaustive outcomes sum to 1.

Complement: $P(\text{not } A) = 1 - P(A)$.

Expected frequency $= nP(A)$.

POSSIBILITY SPACES AND TREES

Use grids/tables for combined equally likely outcomes.

Tree diagrams: multiply along branches; add alternative routes.

Independent events: branch probabilities stay the same.

Dependent events: update probabilities after each outcome.

CONDITIONAL PROBABILITY

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

The denominator is the given/restricted group.

Use two-way tables, Venn diagrams and trees.

Mutually exclusive: $P(A \cap B) = 0$.